

Packet Tracer - Connect Devices using Wireless Technologies

Official Engineering Technical Implementation Report

Lab Introduction & Context

This deployment document provides step-by-step guidance on establishing end-to-end multi-tier network connectivity across various wireless layers. The setup includes an IEEE 802.11 WLAN integration, short-range Bluetooth piconet associations, and redundant smart cellular infrastructure tethering (mobile broadband sharing) using Packet Tracer Physical Mode constraints.

Part 1: Connect a Laptop to the Office WLAN

Concept: Media Interface Adaptation & 802.11 Association

Transitioning from a legacy copper media infrastructure to an airwave RF environment requires swapping out the client physical interface card. Once operational, the interface parses infrastructure **Beacon Frames** to establish sync settings and match security key handshakes (WPA2-PSK) before grabbing a DHCP subnet address.

Step 1: Physical Layer NIC Modification

Opened the target Laptop workspace environment, powered off the internal power rail, extracted the wired PT-LAPTOP-NM-1CFE FastEthernet expansion module, and slotted in the WPC300N wireless antenna array interface card before executing system boot.

Step 2: Employee SSID Association Handshake

Launched the **PC Wireless** desktop utility tool. Initiated airwave inquiries to display the broadcast perimeter, identified the target Employee infrastructure SSID network, and completed security handshake operations using the pre-shared key credentials (Cisco123).

The screenshot displays the 'PC Wireless' desktop utility tool interface. The top navigation bar includes tabs for 'Physical', 'Config' (highlighted), 'Desktop', 'Programming', and 'Attributes'. On the left, a sidebar menu shows 'GLOBAL' (selected) and 'INTERFACE' (with sub-items 'Wireless0' and 'Bluetooth'). The main panel, titled 'Global Settings', contains the following configuration fields:

- Display Name:** Laptop
- Interfaces:** Wireless0 (dropdown menu)
- Gateway/DNS IPv4:**
 - ☒ DHCP
 - ☐ Static
 - Default Gateway:** 192.168.2.1
 - DNS Server:** 203.0.0.2
- Gateway/DNS IPv6:**
 - ☒ Automatic
 - ☐ Static
 - Default Gateway:** [empty field]
 - DNS Server:** [empty field]
- Device Clock:** 00:01:08 Mon Mar 1 1993 UTC
- NTP:**
 - Server Ip:** [empty field]
 - Authentication:**
 - ☒ Enable ☐ Disable
 - Password:** [empty field]

Step 3: Verification & Inband Addressing

Checked the Wireless0 interface properties table to ensure successful initialization of automatic parameters over DHCP, then opened the browser application to resolve the intranet home directory endpoint page at `http://office.srv`.

The screenshot shows the 'Wireless Network Monitor v1.0' application window. It features a green-themed interface with a title bar, a close button, and three tabs: 'Link Information', 'Connect', and 'Profiles'. The 'Link Information' tab is active, displaying a list of available wireless networks. Below the list, there is a 'Site Information' panel showing details for the selected network. To the right of the main content area, there is a '2.4GHz' label and an image of a wireless adapter. At the bottom, a status bar indicates 'Adapter is Active' and provides hardware details.

Below is a list of available wireless networks. To search for more wireless networks, click the **Refresh** button. To view more information about a network, select the wireless network name. To connect to that network, click the **Connect** button below.

Wireless Network Name	CH	Signal
Employee	1	63%

Site Information

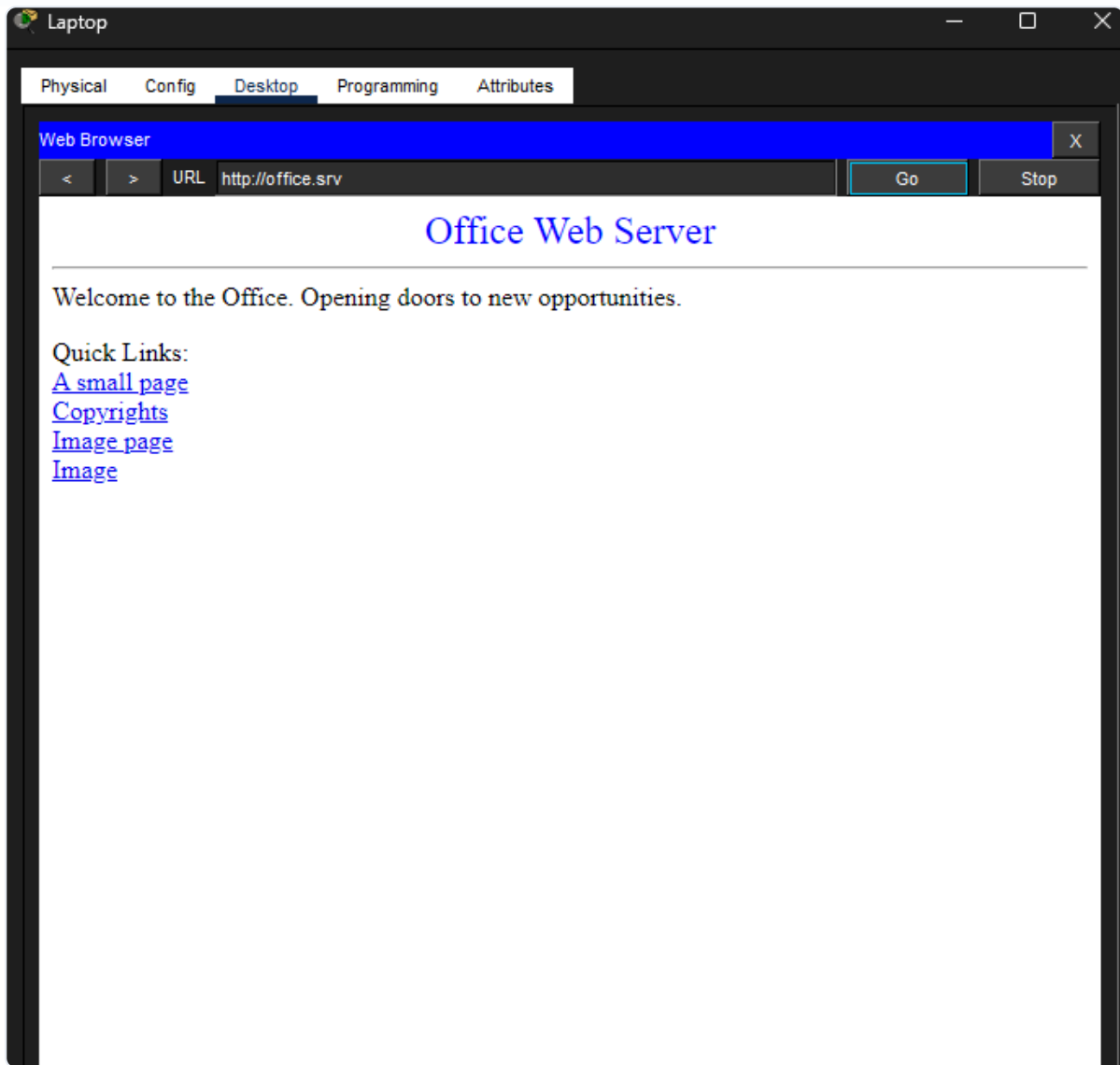
- Wireless Mode: Infrastructure
- Network Type: Mixed B/G/N
- Radio Band: Auto
- Security: WPA2-PSK
- MAC Address: 00D0.BC1C.EA02

Refresh **Connect**

2.4GHz

Adapter is Active

Wireless-N Notebook Adapter Wireless Network Monitor v1.0 Model No. **WPC300N**



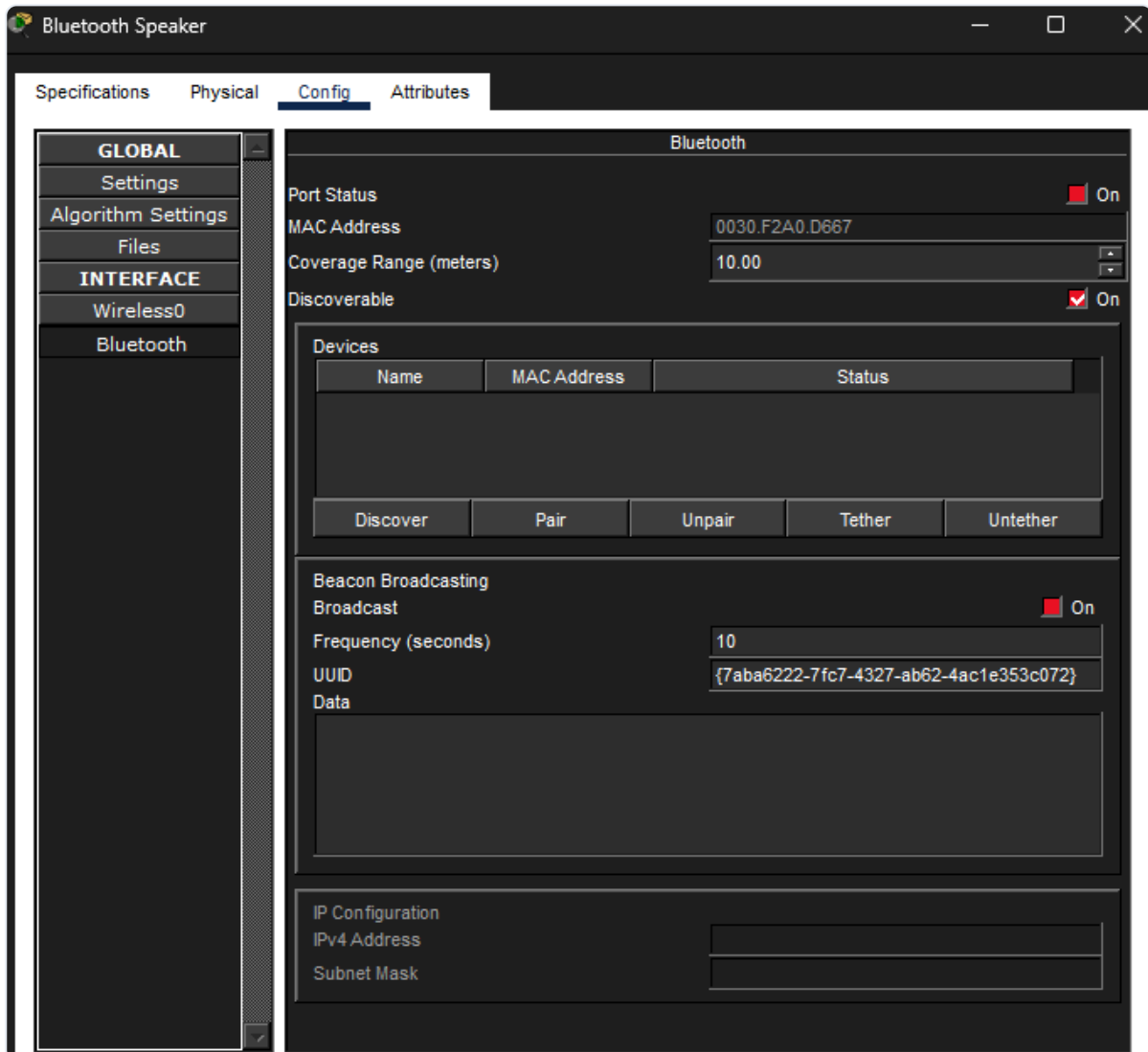
Part 2: Connect Devices with Bluetooth Technology

Concept: Ad-Hoc 802.15.1 WPAN Piconet Mechanics

Bluetooth technology sets up an ad-hoc, short-range **Wireless Personal Area Network (WPAN)** without requiring a centralized base station or wireless controller router. It utilizes a synchronized Master/Slave topology alongside Adaptive Frequency Hopping (AFH) inside the 2.4 GHz ISM band to dodge co-channel interference from nearby high-power enterprise Wi-Fi systems.

Step 1 & 2: Peripheral Device Discovery & Coupling

Activated the integrated Bluetooth module on the standalone Bluetooth Speaker hardware. Opened the Office Tablet settings panel, turned on its corresponding Bluetooth system layer, initiated an active ****Discover**** inquiry scan to identify nearby devices, and coupled the speaker node until status updated to *"Paired, Connected"*. Verified link health by streaming test audio packages using the Music Player app.



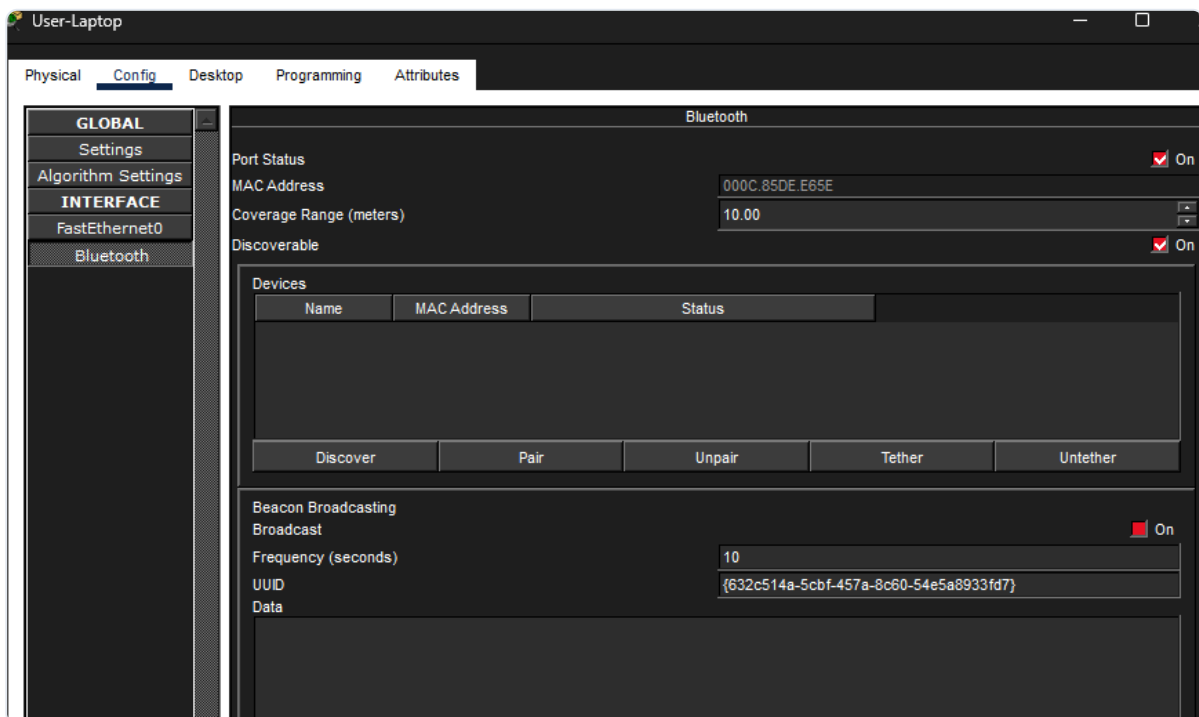
Part 3: Tether a Laptop to Use a Cellular Network via the Smartphone

Concept: Multi-Hop Gateway Overlays & Mobile Hotspots

Tethering turns a standard cellular device into an intelligent personal gateway router. The smartphone maintains its long-range cellular network connection (WAN backhaul link) while simultaneously spinning up a local Bluetooth network interface bridge. This lets independent local clients route traffic directly through it using localized NAT protocols.

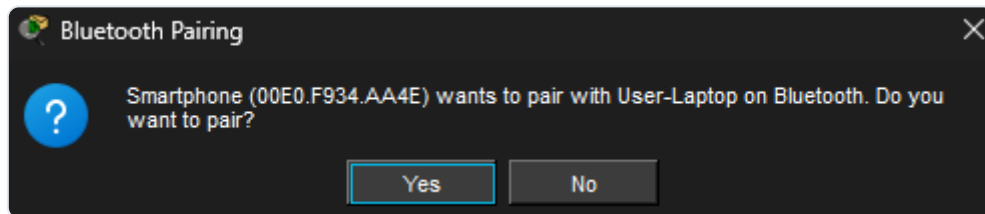
Step 1 & 2: Local Node Setup & WAN Initialization

Activated the Bluetooth radio layer interface on the User-Laptop system. Navigated to the Smartphone configurations panel, toggled the global ****Cellular Tethering**** option on, and checked the 3G/4G Cell11 layout parameters to make sure it grabbed a carrier WAN address from the cellular provider's tower network.

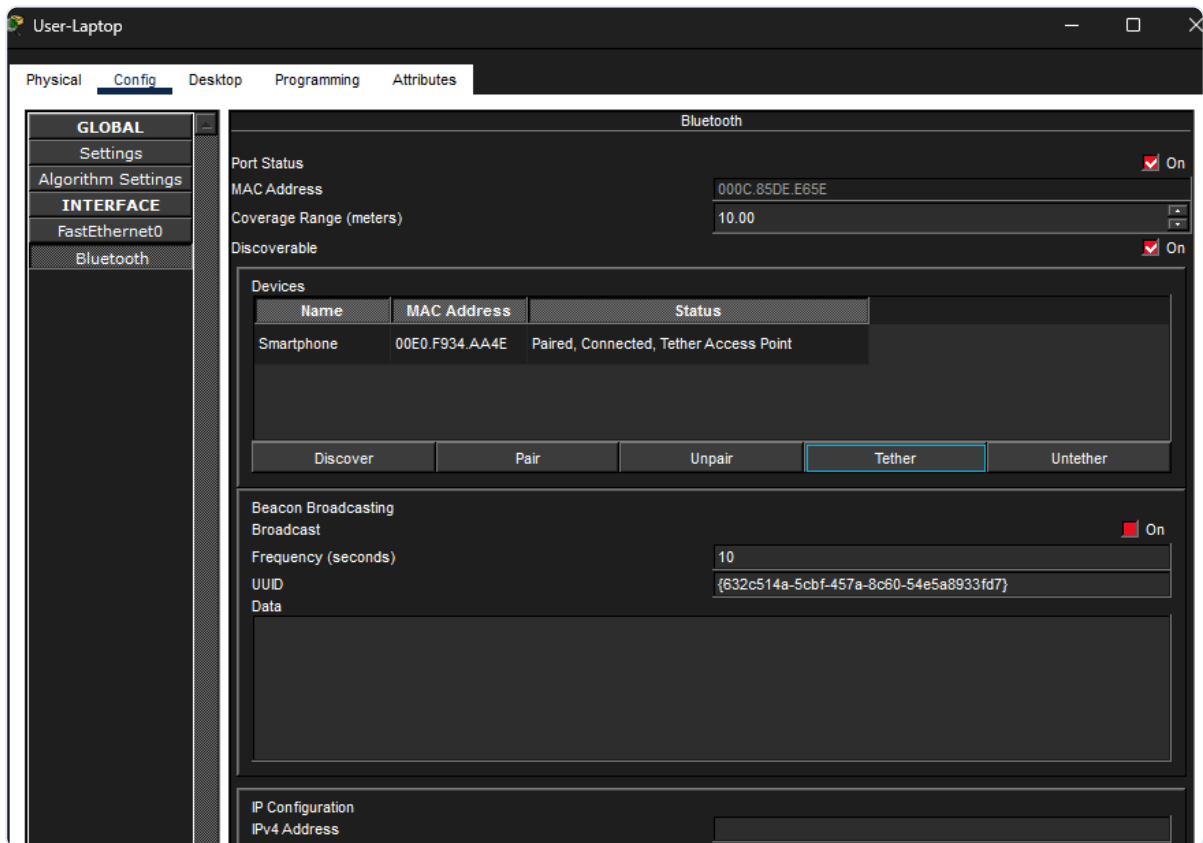


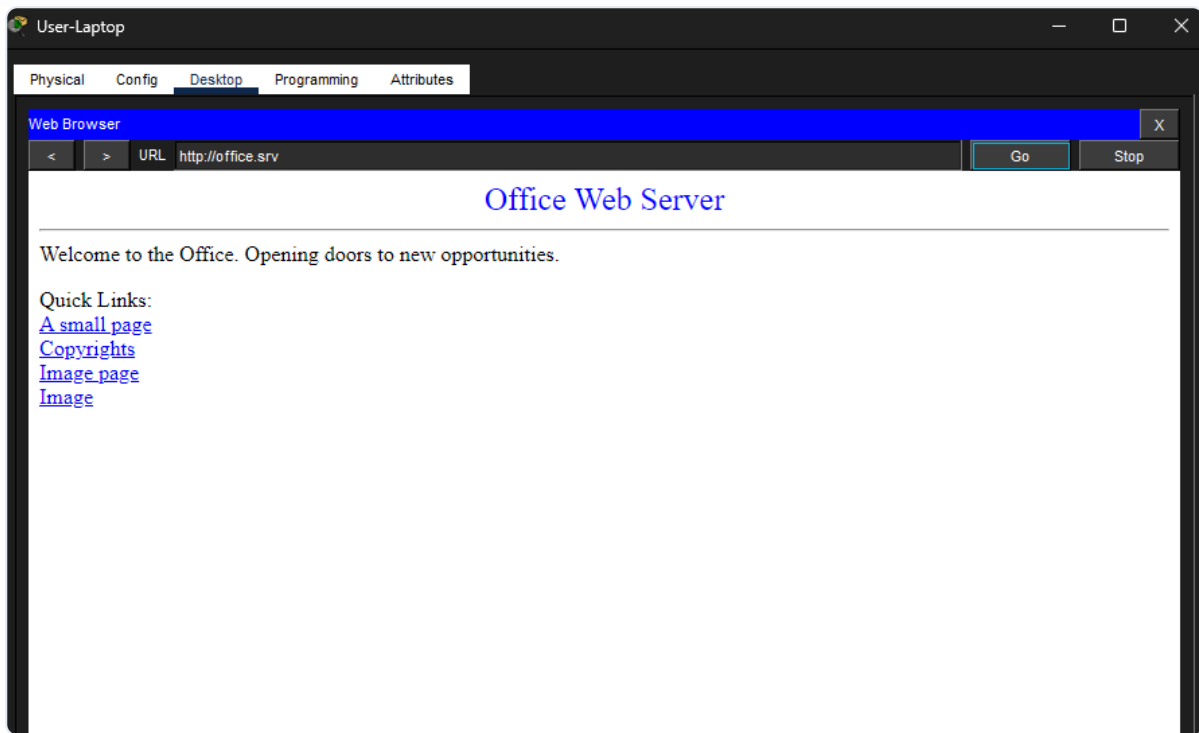
Step 3: Creating the Bluetooth Tether Bridge & Verification

Initiated a Bluetooth search scan from the Smartphone to locate the User-Laptop node, and confirmed the mutual pairing authentication checks. From the laptop's interface configurations table, highlighted the smartphone and clicked ****Tether**** to establish a localized routing path.



Verified that the laptop successfully pulled local address allocations through the phone gateway, then initiated terminal diagnostics via a ping checklist alongside a web browser inquiry test to successfully load up the `http://office.srv` address path.





Part 4: Activity Grade Check

Laboratory Completion Audit

The internal assessment validation routine recorded a 100% completion metrics profile index, verifying that all WLAN interfaces, Bluetooth pairing parameters, and cellular hotspot configurations were completed successfully.

